

Cosmology & the Open Universe

The cosmological layer of the Reconciliation Causal Framework, rewritten for v2.1. Identifies cosmic expansion with the Open Expansion from y_{eff} , states the Friedmann derivation path (Conjecture, not executed), applies the §2.2 dark energy mechanism cosmologically (CONDITIONAL on Friedmann), derives the 120-order resolution, Γ evolution ($(1+z)^{-6}$ EDE suppression), equation of state $w(z) > -1$, and $\Lambda_B = 0$ (Corollary, conditional on T-2). Includes inflation, sector formation, and the Hubble tension reframe. Two inline gap flags (Friedmann execution + dark energy conditional tags).

2 GAPS FLAGGED

7 SUBSECTIONS

Q LAYER

8 CONJECTURES

3 THEOREMS

VERSION

AUDIENCE

DATE

v2.1 Draft (Canonical Form)

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RCF Section 6 — Cosmology

Rewritten Canonical Form — v2.1 (Draft)

Preamble — What This Section Contains and Why

Section 4 constructed the GR perspective (duration, the "now," the burden tensor, Einstein closure). Section 5 examined the saturation regime (black holes). Section 6 applies the framework to the universe as a whole — the cosmological regime where Open Expansion operates at cosmic scale and the dark energy mechanism (established in §2.2) produces the observed acceleration.

What changed from v2.0

v2.0's §6 was "largely intact in architecture" relative to v1.0's §8, with seven targeted corrections. v2.1 retains these and adds the dark-energy conditional tags (M2 from the patch plan):

1. **Friedmann equation honestly labeled** (§6.2, retained from v2.0). v1.0 asserted the Friedmann equation without derivation. v2.0/v2.1 states the 4-step derivation path but acknowledges the computation has not been performed. The $8\pi G/3$ coefficient is imported from standard GR (via κ_B calibration), not derived from RCF primitives.
2. **Dark energy mechanism properly linked to §2.2** (§6.3, retained from v2.0). v1.0 introduced dark energy as "SOE frontier pressure" without connecting it to the restorative drive mechanism. v2.0/v2.1 references the §2.2 derivation ($\rho_{DE} \sim (H/\Gamma_{eff})^2 \cdot \rho_{source}$ with $\Gamma_{eff} \sim \eta \cdot B_{\Delta}$) and applies it cosmologically.
3. **Γ evolution formalized** (§6.3, retained from v2.0). The intuition that Γ was larger in the past (because burden was higher) is given quantitative form: $\rho_{DE} \sim (1+z)^{-3}$, $\Omega_{DE} \sim (1+z)^{-6}$ at recombination.
4. **Cross-reference misattribution fixed** (retained from v2.0). v1.0 cited "Theorem 5.4" for both Einstein closure and singularity avoidance. v2.0/v2.1 disambiguates: Theorem 4.5.4 (Einstein closure) and Theorem 5.2.1 (singularity avoidance).
5. **$\Lambda_B = 0$ downgraded from Theorem to Corollary** (§6.4, retained from v2.0). v1.0 labeled this "Theorem 8.5.1" despite being conditional on T-2. v2.0/v2.1 is a Corollary (of Theorem 4.5.4 / Corollary 1.6.3), honestly conditional.
6. **Q-9 status corrected** (§6.4, retained from v2.0). v1.0's "CLOSED (conditional on T-2)" was self-contradictory. v2.0/v2.1 uses "CONDITIONAL (on T-2)" consistently.
7. **Layer badge corrected** (retained from v2.0). v1.0's "Layers $B \rightarrow C$ with A quarantined" is corrected to Layer Q' (Quartic — cosmological application of coarse-grained hydrodynamics).
8. **Dark energy cosmological predictions CONDITIONAL** (§6.3, NEW in v2.1 — M2 patch). v2.0 labeled the 120-order resolution, EDE suppression, and $w(z)$ as "Theorem." v2.1 adds conditional tags: these are CONDITIONAL on the Friedmann derivation (Conjecture 6.2.2, not yet executed). The structural scaling laws (from §2.2) are Established; the cosmological APPLICATION is conditional.

9. **Equation of state $w(z)$ — falsifiable predictions** (§6.3, retained from v2.0). $w > -1$ always, evolving from 0 (early) to -1 (late). Distinct from both Λ CDM ($w = -1$ exactly) and quintessence.

What this section does NOT contain (deferred)

- The formal action S_{eff} ($\rightarrow$ §7.4)
- The consolidated theorem table ($\rightarrow$ §7.1)
- The 17-point quarantine list ($\rightarrow$ §7.3)

§6.0 Purpose: The Open Universe

Section 6 applies the relational architecture to the universe as a whole. The central identification: cosmic expansion IS Open Expansion at the cosmological frontier. The Hubble rate $H(t)$ is the Open Expansion frontier rate. Dark energy is the restorative drive (from §2.2) applied at cosmological scale — the residual burden of global desynchronization (Theorem 1.6.4 / Theorem 4.2.3).

The key sequence is:

1. **Expansion = Open Expansion frontier rate** (§6.1) — $H(t) = \gamma_{\text{eff}}(t)$.
2. **Friedmann-like dynamics** (§6.2) — Conjecture, derivation path stated but not executed.
3. **Dark energy cosmological application** (§6.3) — $\rho_{\text{DE}} \sim (H/\Gamma)^2$, 120-order resolution, Γ evolution, EDE suppression, $w(z)$. CONDITIONAL on §6.2.
4. **$\Lambda_B = 0$** (§6.4) — Corollary, conditional on T-2.
5. **Early universe** (§6.5) — initial state, inflation, sector formation.
6. **Status and guardrails** (§6.6).

§6.1 Expansion as Open Expansion

Conjecture 6.1.1 (Hubble Rate = Open Expansion Frontier Rate)

The Hubble expansion rate is the Open Expansion frontier rate:

$$H(t) = \gamma_{\text{eff}}(t) = \frac{1}{V} \cdot \frac{dV}{dt}$$

The universe expands because new Open Expansions continuously incorporate new events at speed $c = \gamma \cdot \square_0$ (Theorem 1.1.4). The expansion rate is the rate at which the causal frontier advances.

Status: Structural conjecture. The identification $H = \gamma_{\text{eff}}$ is physically motivated (both measure the rate of volume growth) but not formally derived from the FLRW reduction of the ADM ansatz (which is Conjecture 6.2.2, not yet executed).

Corollary 6.1.2 (Expansion Rate Limit)

The Hubble rate is bounded:

$$H(t) \leq \gamma_{\max} = \frac{1}{\varepsilon} = \frac{\gamma}{\ell_0}$$

where $\varepsilon = 1/\gamma$ is the fundamental time step. The expansion cannot exceed the rate at which Open Expansion steps complete.

Status: Conditional on Conjecture 6.1.1 and Theorem 1.1.4 ($\gamma = \sqrt{\lambda_{\min}(F)}$).

Conjecture 6.1.3 (Cubic Volume Growth)

The cubic correlation volume V_ω (Definition 3.6.6) grows with the scale factor:

$$\mathcal{V}_\omega(t) \propto a(t)^3$$

where $a(t)$ is the cosmological scale factor. This connects the algebraic correlation volume to the FLRW scale factor.

Status: Structural conjecture. The connection between the emergent geometry's volume element and the FLRW scale factor requires the Friedmann derivation (§6.2).

§6.2 Friedmann-Like Dynamics

Conjecture 6.2.1 (FLRW Reduction of the ADM Ansatz)

The burden-metric ADM ansatz (Definition 4.5.1), restricted to homogeneous and isotropic configurations (FLRW family), takes the form:

$$ds^2 = -N_B^2 dt^2 + a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right)$$

where $N_B = \alpha(B)$ is the burden-clock lapse (§4.1.1), $a(t)$ is the scale factor, and k is the spatial curvature.

Status: Structural conjecture. The restriction to FLRW requires the burden tensor to be homogeneous and isotropic — plausible but not proven.

Conjecture 6.2.2 (Friedmann Equation — DERIVATION PATH STATED, NOT EXECUTED)

The Friedmann-like equation is the Euler-Lagrange equation of Open Extension descent on the FLRW metric:

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho_B - \frac{k}{a^2}$$

where ρ_B is the burden density.

The 4-step derivation path (stated but not executed):

1. **FLRW-restricted ADM ansatz:** restrict the burden-metric ADM ansatz (Definition 4.5.1) to the FLRW family (Conjecture 6.2.1).
2. **Homogeneous/isotropic burden tensor components:** compute $^{(B)}_{\mu\nu}$ for the FLRW metric — the burden density ρ_B and pressure p_B .
3. **S_{eff} variation:** vary S_{eff} (§7.4) with respect to $a(t)$ to obtain the Euler-Lagrange equation.

4. **κ_B calibration**: use $\kappa_B = 8\pi G/c^4$ (calibrated by the Newtonian limit, Theorem 4.6.1) to fix the $8\pi G/3$ coefficient.

Status: [Structural conjecture — derivation path stated, computation NOT performed]. Two specific gaps:

(a) The $8\pi G/3$ coefficient is **imported from standard GR** (via the κ_B calibration), NOT derived from RCF primitives. The calibration gives $\kappa_B = 8\pi G/c^4$; the $1/3$ factor comes from the FLRW-specific reduction of the Einstein equation, which is standard but not yet performed within RCF.

(b) The non-local relational channel $\wedge(\text{rel})$ (built from Π^{net} , §3.9) requires FLRW averaging that v1.0 never addressed and v2.0/v2.1 leaves as part of the open derivation.

[GAP: v2.1 — the Friedmann derivation (Conjecture 6.2.2) is the M5 fix from the patch plan (severity MATERIAL, blocks Phase C). The 4-step path is stated but not executed. Until the FLRW ADM variation is performed, the Friedmann equation remains a structural conjecture, and all cosmological predictions in §6.3 inherit this conditional status. Flagged as GAP-6.2.2, severity MATERIAL (blocks Q-13 Λ CDM recovery).]

Remark 6.2.3 (Consolidated forward references)

Conjecture 6.2.2 resolves the following forward references:

- The $\Lambda_B = 0$ forward reference from §4.5.5 (resolved by Corollary 6.4.1).
- The dark energy reinterpretation from §2.2 (resolved by Conjecture 6.3.2).
- The cubic volume $\leftrightarrow$ scale factor connection from §6.1.3 (resolved by the FLRW reduction).

§6.3 The Dark Energy Cosmological Application

Conjecture 6.3.1 (Dark Sector Decomposition)

The dark sector decomposes into three parts:

1. **Primary Dark Matter** (relational burden): $\wedge(\text{rel}) = [\hat{C}_\alpha, \Pi^{\text{net}}]$ (Theorem 4.4.3). This is the DERIVED dark-matter mechanism — cross-sector linking cost. Q-1 (amplitudes) OPEN — requires numerical simulation.
2. **Cross-sector gravity**: QUARANTINED (crosses the FIREWALL, §2.8.3). Not a derived mechanism.
3. **Dark Energy** (residual burden pressure): the restorative drive from §2.2, applied cosmologically. NOT a cosmological constant ($\Lambda_B = 0$, Corollary 6.4.1).

Conjecture 6.3.2 (Dark Energy Density — Cosmological Application)

[v2.1: CONDITIONAL on the Friedmann derivation (Conjecture 6.2.2). The structural scaling from §2.2 is Established; the cosmological application is conditional.]

The dark energy density scales as:

$$\rho_{\text{DE}}(t) \sim \left(\frac{H(t)}{\Gamma_{\text{eff}}(t)} \right)^2 \cdot \rho_{\text{source}}$$

where:

- $H(t)$ is the Hubble rate (Conjecture 6.1.1),

- $\Gamma_{\text{eff}}(t) \sim \eta \cdot B_{\Delta}(t)$ is the burden-dependent relaxation rate (Property 2.2.2),
- ρ_{source} is the source burden density.

The quadratic scaling $(H/\Gamma)^2$ comes from burden being quadratic in mismatch ($B_{\Delta} = |\text{mismatch}|^2$, Remark 1.3.2). The exponent 2 is NOT tuned — it is structural.

Status: The structural scaling $\rho_{\text{DE}} \sim (H/\Gamma)^2$ is Established (from §2.2). The cosmological application — identifying H with the observed Hubble rate and comparing ρ_{DE} to observational data — is CONDITIONAL on the Friedmann derivation (Conjecture 6.2.2).

[GAP: v2.1 — the dark energy cosmological predictions (Conjectures 6.3.2–6.3.5) are CONDITIONAL on the Friedmann derivation (GAP-6.2.2). This is the M2 fix from the patch plan: the structural mechanism is established, but the cosmological predictions inherit the Friedmann's conditional status. Flagged as GAP-6.3.2, severity MATERIAL (propagated from GAP-6.2.2).]

Theorem 6.3.3 (120-Order Resolution)

[v2.1: Structural scaling ESTABLISHED; cosmological identification CONDITIONAL on §6.2.]

The dark energy density relative to the Planck density is:

$$\frac{\rho_{\text{DE}}}{\rho_{\text{Planck}}} \sim \left(\frac{H_0}{\Gamma} \right)^2 \sim (H_0 \cdot t_P)^2$$

This is the observed value. The 120-order suppression is $(H_0/\Gamma)^2 = (\Omega_P/R_H)^2$ — the square of the IR/UV scale ratio. The exponent 2 is not tuned; it comes from burden being quadratic in mismatch.

Status: The structural scaling is Established (from the burden-quadratic structure, §2.2). The identification with the observed 10^{-122} is CONDITIONAL on the Friedmann derivation (§6.2) and the Hubble rate identification (Conjecture 6.1.1).

Theorem 6.3.4 (Γ Evolution and Early Dark Energy Suppression)

[v2.1: Structural scaling ESTABLISHED; cosmological identification CONDITIONAL on §6.2.]

$\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}$ was larger in the past (because B_{Δ} was larger — higher burden, steeper gradient). In matter domination ($H \sim (1+z)^{3/2}$, $B_{\Delta} \sim \rho_{\text{matter}} \sim (1+z)^3$):

$$\rho_{\text{DE}} \sim (1+z)^{-3}, \quad \Omega_{\text{DE}} \sim (1+z)^{-6}$$

giving $\Omega_{\text{DE}} \sim 10^{-18}$ at recombination ($z \approx 1100$) — eighteen orders of magnitude below the CMB bound ($\Omega_{\text{EDE}} < 0.004$). Early dark energy tension vanishes.

Status: The structural scaling is Established. The cosmological application is CONDITIONAL on §6.2.

Theorem 6.3.5 (Equation of State — Falsifiable Predictions)

[v2.1: Structural scaling ESTABLISHED; cosmological identification CONDITIONAL on §6.2.]

The equation of state evolves:

$$w(t) = -1 - \frac{2\dot{H}}{3H^2}$$

Predictions (distinct from both Λ CDM and quintessence):

1. $w > -1$ always (no phantom crossing).

2. w evolves from 0 (early, matter domination) to -1 (late time).
3. $\rho_{DE} \rightarrow 0$ as $t \rightarrow \infty$ (decays, not constant).

These are testable against DESI/Euclid and Nancy Grace Roman.

Status: The structural derivation (from $\rho_{DE} \sim H^2$) is Established. The cosmological identification and observational comparison are CONDITIONAL on §6.2.

Corollary 6.3.6 (Asymptotic Decay — UPGRADED)

$$\lim_{t \rightarrow \infty} \rho_{DE}(t) = 0$$

Status: The asymptotic decay follows from the Convergence Theorem (Theorem 1.6.1) applied at global scale — the universe asymptotically approaches a flat Master-Zero state. v1.0's "Conjecture 8.3.3" is UPGRADED: the asymptotic decay is established (from the Convergence Theorem); only the present-day amplitude requires T-2.

§6.4 $\Lambda_B = 0$

Corollary 6.4.1 ($\Lambda_B = 0$ — Corollary, NOT Theorem)

[Retained from v2.0. v1.0's "Theorem 8.5.1" is downgraded to Corollary — a theorem conditional on an unproved conjecture is not a theorem.]

The bare cosmological constant vanishes:

$$\Lambda_B = 0$$

Proof. $\Lambda_B = 0$ because the Master-Zero condition $\omega(M^*) = 0$ is an exact asymptotic attractor (Corollary 1.6.3), not a tuned parameter. There is no vacuum energy to fine-tune.

Status: [Conditional on T-2]. This is a COROLLARY of Theorem 4.5.4 (Einstein closure) and Corollary 1.6.3 ($\Lambda_B = 0$ from Master-Zero). The v1.0 label "Theorem 8.5.1" was inconsistent with its T-2 dependency — a theorem conditional on an unproved conjecture is not a theorem; it is a conditional proposition. v2.0/v2.1 corrects this to a Corollary, honestly conditional.

Remark 6.4.2 (Q-9 Status Corrected)

[Retained from v2.0.]

Q-9 (the cosmological constant problem) status:

- The 120-order discrepancy is STRUCTURALLY RESOLVED (Theorem 6.3.3 — no vacuum energy, the observed ρ_{DE} is a dynamic restorative drive).
- The formal closure $\Lambda_B = 0$ is CONDITIONAL on T-2 (Corollary 6.4.1).

The v1.0 label "CLOSED (conditional on T-2)" was self-contradictory — "CLOSED" and "conditional on an open target" cannot both hold. v2.0/v2.1 uses "CONDITIONAL (on T-2)" consistently.

§6.5 The Early Universe

Conjecture 6.5.1 (Initial State — Maximal Burden)

The universe began in a state of maximal burden:

$$B_{\Delta}(t \rightarrow 0) \rightarrow B_{\Delta}^{\max}$$

This is the boundary condition, not a derived result. The initial state is the maximally fractured, maximally desynchronized composite — the starting point from which Open Expansion and Open Extension drive the system toward Master-Zero.

Conjecture 6.5.2 (Inflation = Rapid Open Expansion at $\gamma_{\max}$)

Cosmic inflation was the initial phase of maximal Open Expansion at $\gamma_{\max}$ (Corollary 6.1.2), with Open Extension descent suppressing cross-sub-sector coherence:

$$H(t_{\text{inflation}}) \approx \gamma_{\max} = \frac{\gamma}{\ell_{\theta}}$$

Status: Q-3 OPEN — separate from the core framework per Spec Ch. 12.2. The inflationary observables (scalar/tensor perturbation spectra) require a model of the Open Expansion frontier fluctuation spectrum, which is not yet developed.

Conjecture 6.5.3 (Sector Formation Epochs)

Record sectors (§2.6) formed as Open Extension descent crossed decoherence thresholds:

$$B_{\text{cross}}(i, j) \gg B_{\text{intra}}(i) \quad \text{at } t_{\text{formation}}$$

Status: Q-4 PARTIAL. T-4 (Born rule) is STRENGTHENED (§2.7.5), but the amplitude derivation (the specific p_k values at sector formation) is open.

§6.6 Status and Guardrails

Remark 6.6.1 (Q-13 Λ CDM Recovery Status)

Q-13 (Λ CDM recovery) status: PARTIAL.

- FLRW form: Conjecture 6.2.2 (derivation path stated, not executed — GAP-6.2.2).
- Dark sector mechanisms: established (relational burden = DM, restorative drive = DE).
- Amplitudes ($H_0, \Omega_m, \Omega_{DM}, \Omega_{\Lambda}, \text{eff}$): OPEN — require numerical simulation of Open Extension descent on FLRW.

Remark 6.6.2 (Hubble Tension — Structural Reframe)

[Retained from v2.0/v2.1.]

The Hubble tension (H_0 discrepancy between CMB-inferred and locally-measured values) may reflect local B_{Δ} variation:

$$H_0^{\text{local}} \neq H_0^{\text{global}} \quad \text{because} \quad \gamma_{\text{eff}}^{\text{local}} \neq \gamma_{\text{eff}}^{\text{global}}$$

Local Open Extension rate may differ from the global average inferred from the CMB. This is a PREDICTION (not derived) — if the Hubble tension is due to local burden variation, it should correlate with local matter density.

Guardrail 6.6.3 (Eight Cosmological Guardrails)

1. $\Lambda_B = 0$ is conditional on T-2, not EXACT.
2. The Friedmann equation is a Conjecture, not a Theorem.
3. Dark energy cosmological predictions are CONDITIONAL on the Friedmann derivation.
4. The $8\pi G/3$ coefficient is imported from standard GR, not derived from RCF primitives.
5. The non-local relational channel $\wedge(\text{rel})$ requires FLRW averaging (not yet performed).
6. Inflation (Q-3) is separate from the core framework.
7. The Hubble tension reframe is a prediction, not a derivation.
8. The asymptotic decay $\rho_{\text{DE}} \rightarrow 0$ is established (from the Convergence Theorem); the present-day amplitude is T-2-conditional.

§6.7 Architectural Summary

What Section 6 Contains

Subsection	Content	Layer
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§6.0	Purpose: the open universe	—
§6.1	$H(t) = \gamma_{\text{eff}}$, expansion rate limit, cubic volume growth	Q'
§6.2	Friedmann equation (Conjecture, derivation path stated, not executed)	Q'
§6.3	Dark energy application (ρ_{DE} , 120-order, Γ evolution, EDE suppression, $w(z)$) — CONDITIONAL on §6.2	Q'
§6.4	$\Lambda_B = 0$ (Corollary, conditional on T-2), Q-9 corrected	Q'
§6.5	Initial state, inflation, sector formation	Q'
§6.6	Q-13 status, Hubble tension, eight guardrails	Q'

Acyclicity Test

Verification:

- §6.1 uses γ (§1.1.4), ℓ_0 (§0.3.6), volume element (§3.6.6). ✓

- §6.2 uses ADM (§4.5.1), burden tensor (§4.3), κ_B (§4.5.6). ✓
- §6.3 uses dark energy mechanism (§2.2), B_Δ (§1.3), H (§6.1.1). ✓
- §6.4 uses $\Lambda_B = 0$ (§1.6.3, §4.5.5). ✓
- §6.5 uses B_Δ , $\gamma_{\max}$ (§6.1.2), record sectors (§2.6). ✓
- §6.6 references all prior. ✓

Verdict: Section 6 is acyclic.

Inline Gap Summary for §6

Gap ID	Location	Description	Severity	Blocks
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GAP-6.2.2	§6.2 Conjecture 6.2.2	Friedmann derivation path stated, not executed (M5)	MATERIAL	Q-13 Λ CDM recovery
GAP-6.3.2	§6.3 Conjecture 6.3.2	Dark energy cosmological predictions conditional on Friedmann (M2)	MATERIAL	Observational comparison

Total gaps in §6: 2 (0 fatal, 2 material). Both are pre-existing from the patch plan (M5, M2). No NEW structural gaps introduced.

Section 6 is CLOSED. Section 7 (Audit, Formal Action, and Closure) can be merged against this stable cosmological framework.